CAT

Analysis Content and Results

Business intelligence, data engineering, analytics and predictive modeling results

Project	End-to-End Business Intelligence and Predictive Analytics Solution using Databricks and Power BI
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Data source	Google BigQuery public dataset - TheLook E-Commerce
Technology	Google BigQuery -> Databricks (Bronze/Silver/Gold + ML) -> Power BI

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1. Executive Summary

This project implements a complete business intelligence and predictive analytics workflow for an e-commerce scenario. The solution begins with a public Google BigQuery dataset, moves the data into Databricks, applies a Bronze-Silver-Gold architecture, creates an analytical star schema, builds a machine-learning model for second-purchase propensity, and finally exposes business KPIs and predictive outputs in a six-page Power BI report.

End-to-end flow: BigQuery -> CSV staging -> Databricks Bronze -> Silver -> Gold -> Machine Learning -> Power BI

The analytical layer contains 100,000 users, 125,545 orders, 29,120 products and 182,235 validated order-item rows. The dashboard reports approximately EUR 10.86M revenue, EUR 5.64M profit and a 51.9% profit margin in the full report view. The final predictive model is Logistic Regression with ROC-AUC 0.697, PR-AUC 0.334 and recall 76.3% on the chronological final test period.

Revenue	Profit	Profit Margin	Orders
10.86M	**5.64M**	**51.9%**	**125,545**
Customers	Gold Items	Test ROC-AUC	Test Recall
~80K in report	**182,235**	**0.697**	**76.3%**

2. Business Questions and Scope

The project addresses two connected needs: descriptive business intelligence and forward-looking customer prioritization.

- How much revenue and profit does the business generate, and how do they change over time?
- Which traffic sources, countries, categories, brands and products contribute most to activity?
- How are orders distributed across Shipped, Complete, Processing, Cancelled and Returned statuses?
- What are the return and cancellation rates, and how do they evolve over time?
- Can a first-time customer be ranked by likelihood of making a second non-cancelled purchase within 180 days?

The final ML business question was deliberately narrowed to the first-to-second purchase transition because the synthetic TheLook dataset has an artificial ceiling on orders per customer. This prevents the model from learning a dataset artifact instead of useful behavior.

3. Data Source and Analytical Architecture

3.1 Source data

The source is the Google BigQuery public dataset TheLook E-Commerce (bigquery-public-data.thelook_ecommerce). Four core tables were used: users, orders, order_items and products. The tables were exported as CSV files; the larger datasets were stored in multiple CSV parts and uploaded into a Databricks Volume at /Volumes/workspace/default/bi_final_raw/.

Source table	Bronze rows	Role
users	100,000	Customer demographics, geography and acquisition source
orders	125,545	Order header, lifecycle status and timestamps
order_items	182,295	Item-level transactions and sale prices
products	29,120	Product, brand, category, cost and retail attributes

3.2 Medallion architecture

Layer	Purpose	Main output

Raw / Volume	Preserve exported CSV input files	bi_final_raw
Bronze	Load source structure with minimal transformation; add ingestion timestamp	bronze_users, bronze_orders, bronze_order_items, bronze_products
Silver	Validate keys, duplicates, nulls and referential integrity; remove invalid transaction rows	silver_users, silver_orders, silver_order_items, silver_products
Gold	Create BI-ready dimensions/facts and calculated financial fields	dim_users, dim_products, dim_date, fact_orders, fact_order_items
ML outputs	Expose current scores, test predictions and model metrics	ml_second_purchase_predictions, ml_second_purchase_test_predictions, ml_second_purchase_model_metrics
Power BI	Semantic model, DAX measures and six analytical report pages	Interactive BI report

4. Data Quality and Warehouse Results

4.1 Data quality findings

Primary-key checks showed no duplicate user IDs, order IDs, order-item IDs or product IDs. Referential integrity checks found no orders without a valid user and no order items without a matching product. However, 60 order-item rows referenced 43 order IDs that were absent from the orders table.

Duplicate Users	Duplicate Orders	Duplicate Items	Duplicate Products
0	0	0	0
Orders without User	**Items without Order**	**Missing Order IDs**	**Items without Product**
0	60	43	0

The 60 orphan order-item rows were removed from the analytical transaction layer by inner-joining order_items to valid orders. This reduced order_items from 182,295 Bronze rows to 182,235 Silver rows while retaining all 100,000 users, 125,545 orders and 29,120 products.

4.2 Gold analytical model

Gold table	Rows	Purpose
dim_users	100,000	Customer dimension
dim_products	29,120	Product dimension
dim_date	2,821	Calendar dimension
fact_orders	125,545	Order header fact
fact_order_items	182,235	Item-level revenue/cost/profit fact

Revenue was defined as sale_price. Product cost was joined from the product dimension and profit was calculated as sale_price - product_cost. A continuous date dimension was generated for the full order-date range and enriched with year, month, month name, quarter, day, day name and year-month.

5. Business Intelligence Findings

5.1 Executive KPIs

Total Revenue	Total Profit	Profit Margin	Total Orders
EUR 10.86M	EUR 5.64M	51.9%	126K
Customers	**Average Order Value**		
80K	86.52		

The Executive Overview shows revenue and profit at a scale large enough to support category, brand, country and traffic-source comparisons. Search is the dominant traffic source in the report, and China is the largest country by

revenue, followed by the United States and Brazil. The order-status mix is led by Shipped, followed by Complete, Processing, Cancelled and Returned.

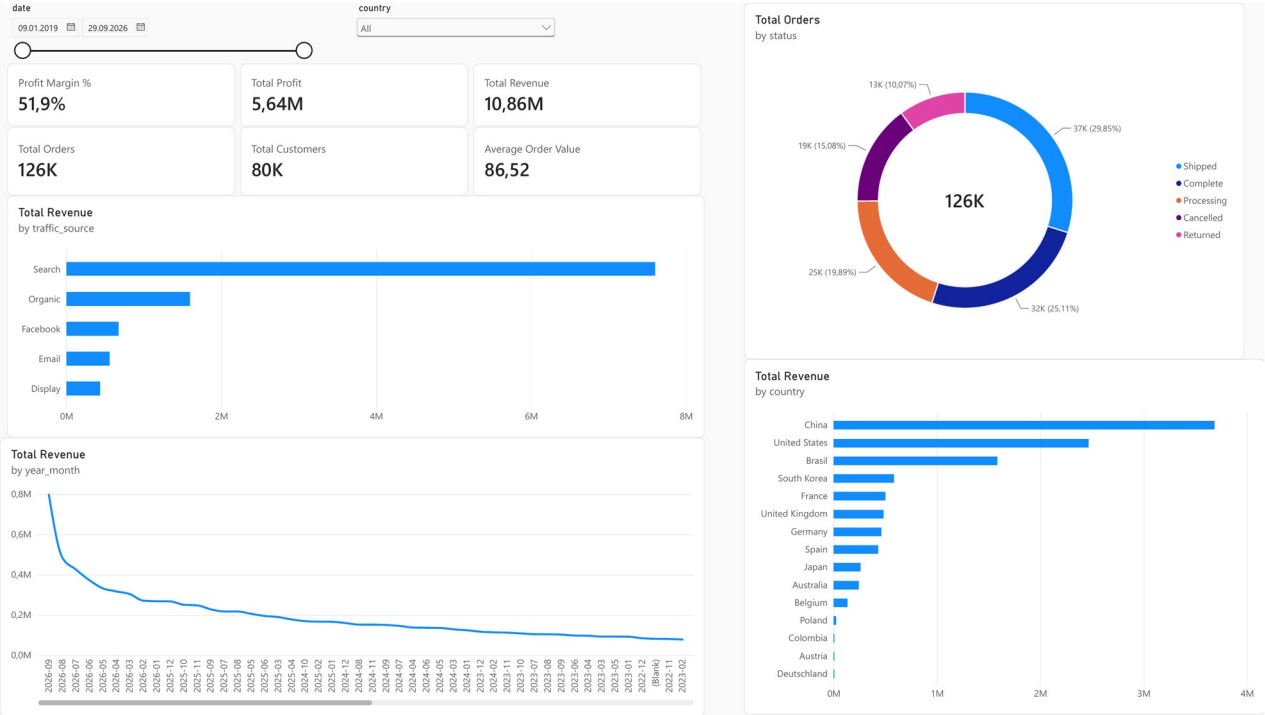


Figure 1. Executive Overview - KPIs, revenue trend, traffic source, order status and country performance.

5.2 Sales, profit and product performance

The Sales & Profit page compares both revenue and profit over time and across product categories and brands. Outerwear & Coats and Jeans are among the strongest categories in the saved report view. The product page then drills down to top products and uses a revenue-versus-profit scatter plot to reveal the strong positive relationship between sales volume and profit at product level.

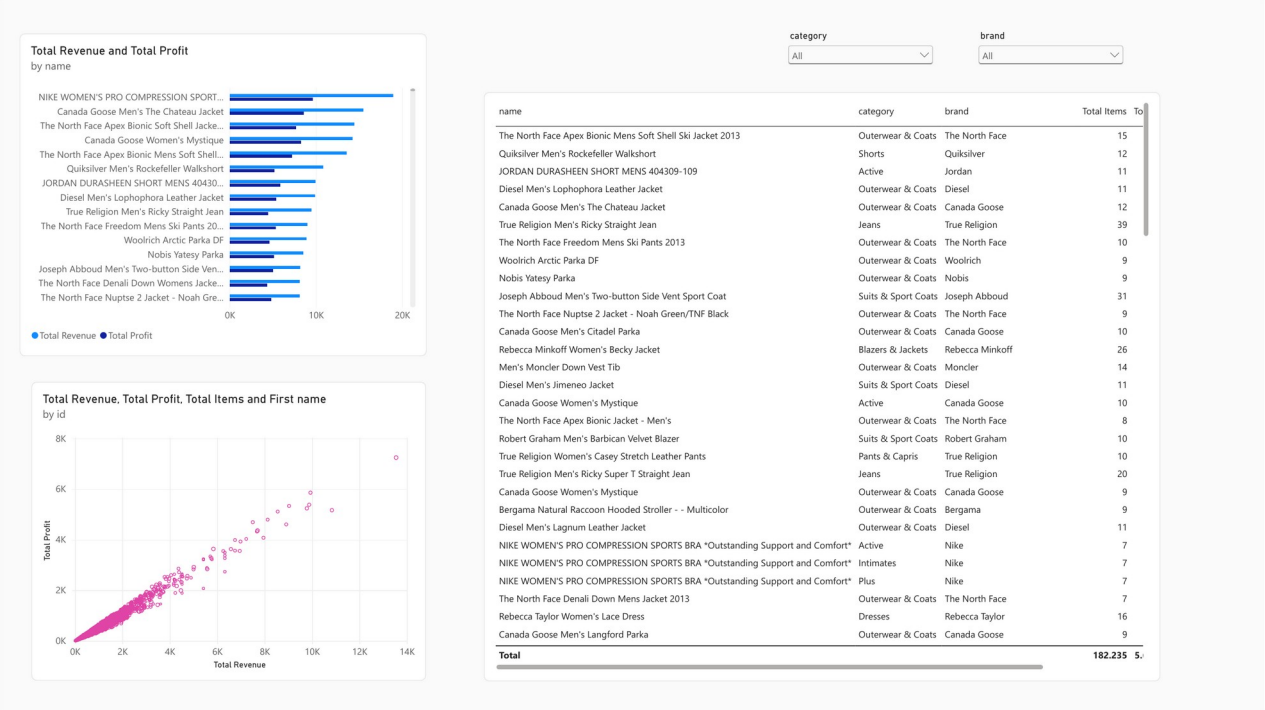


Figure 2. Product Performance - top products, revenue-profit scatter plot and detailed product table.

5.3 Orders, returns and cancellations

The report contains 125,545 orders. In the saved full report view, approximately 15.1% of orders are Cancelled and 10.1% are Returned. The page combines KPI cards, order-status composition, a time trend for returned/cancelled orders, and a country-by-status comparison.

Shipped	Complete	Processing	Cancelled
37,479	**31,525**	**24,973**	**18,931**
Returned	Cancellation Rate	Return Rate	
12,637	**15.1%**	**10.1%**	

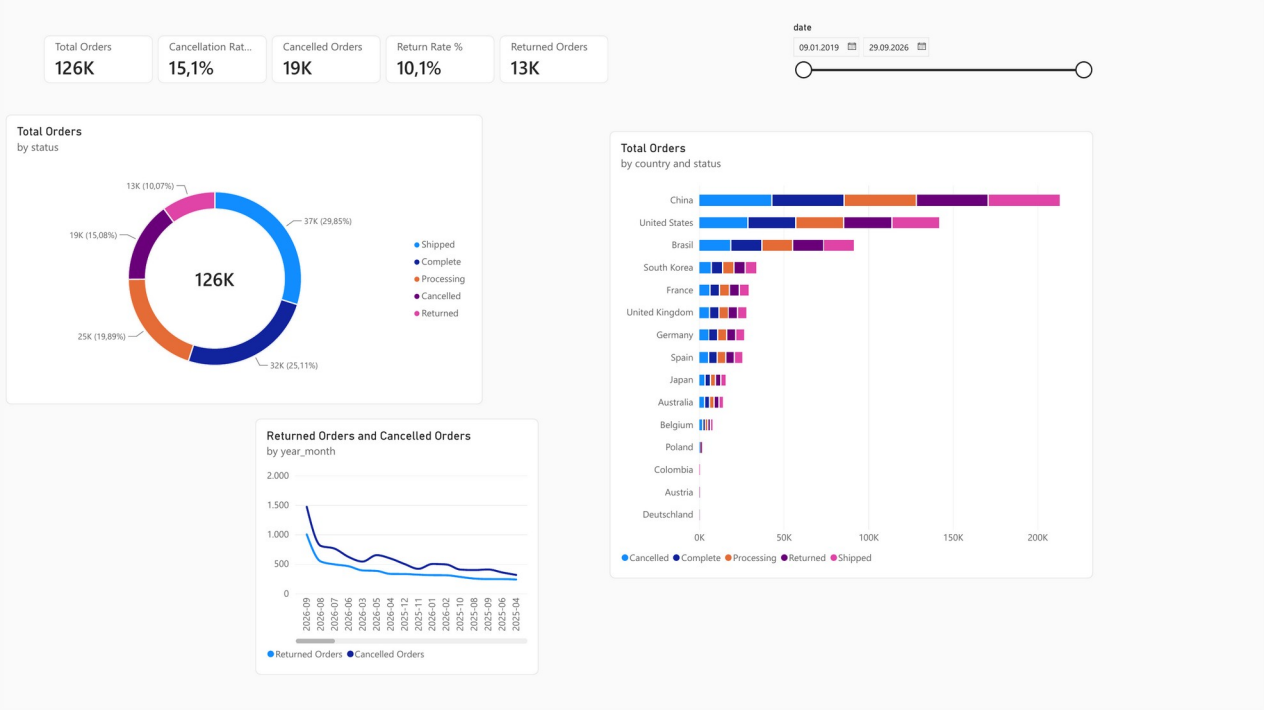


Figure 3. Orders & Returns - status mix, cancellation/return KPIs, time trend and country comparison.

6. Machine Learning Analysis and Results

6.1 Target definition and leakage control

A purchase is defined as a non-cancelled order. For every first purchase, the target equals 1 when a second non-cancelled purchase occurs within 180 days, otherwise 0. First orders without a complete 180-day observation window are excluded from labeled modeling data. Only variables known at or before the first order are used as features, preventing future leakage.

Cohort / split	Rows	Positive rate	Time range
Mature first-order cohort	56,570	11.48%	First order <= 02 Apr 2026
Train	39,602	8.70%	09 Jan 2019 - 17 Apr 2025
Validation	8,484	15.30%	17 Apr 2025 - 27 Oct 2025
Final Test	8,484	20.63%	27 Oct 2025 - 02 Apr 2026

The changing target rate across time is itself an important analytical finding: repurchase prevalence rises substantially from Train to Validation to Test, demonstrating temporal drift. Therefore the model was evaluated chronologically rather than with a random split.

6.2 Feature engineering and model candidates

Features included first-order basket size, order value, average item price, cost, gross margin, distinct products/categories/brands, customer tenure, age, cyclical time features, weekend indicator, gender, country and traffic source. Identifiers and future outcome fields were explicitly excluded. Numeric missing values were imputed; categorical variables were indexed and one-hot encoded. Class weights addressed the imbalanced target.

Candidate	Validation threshold	ROC-AUC	PR-AUC	Precision	Recall	F1
Logistic Regression	0.50	0.660	0.231	23.4%	62.2%	0.341
Random Forest	0.45	0.643	0.225	22.8%	61.9%	0.333
Gradient-Boosted Trees	0.45	0.618	0.211	20.8%	60.3%	0.309
No-skill prevalence	-	0.500	0.153	-	-	-

Logistic Regression was selected because it delivered the best Validation PR-AUC while remaining straightforward to interpret. The decision threshold was locked on Validation at 0.50 before the final Test evaluation.

6.3 Final test results

ROC-AUC	PR-AUC	Precision	Recall
0.697	**0.334**	**30.6%**	**76.3%**
F1	Accuracy	Specificity	Brier Score
0.436	**59.3%**	**54.9%**	**0.221**

	Predicted No	Predicted Yes
Actual No	3,700	3,034
Actual Yes	415	1,335

The model identifies 1,335 of 1,750 actual repurchasers in the final test period. PR-AUC (0.334) is materially above the no-skill baseline of 0.206, which is more informative than accuracy for this imbalanced outcome.

6.4 Ranking quality and interpretation

The main business value is ranking rather than treating the output as a perfectly calibrated probability. The lowest test-score decile has an observed repurchase rate of about 6.0%, while the highest decile reaches about 37.7% - more than six times higher. The strongest signed model effect is customer tenure at first order (negative coefficient), while other individual effects are much smaller. These are associations in synthetic data, not causal effects.

The model scored 12,235 current first-time customers who had not yet repurchased and whose complete 180-day outcome was not yet known. Quantile cutoffs around 0.491 and 0.577 created Low, Medium and High Repurchase Probability segments for business prioritization.

7. Power BI Reporting Output

Power BI imports the Gold analytical tables plus the current ML predictions and model metrics. The semantic model uses shared dimensions to filter multiple fact/prediction tables and intentionally avoids a direct fact_orders-to-fact_order_items relationship to prevent ambiguous filtering and double counting.

Report page	Purpose
01 Executive Overview	Revenue, profit, orders, customers, AOV, traffic source, status, country and time trend
02 Sales & Profit	Revenue/profit trends, category and brand performance
03 Customer & Marketing	Customer count, revenue per customer, age, gender, traffic source and country
04 Product Performance	Top products, product-level revenue/profit relationship and detail table

05 Orders & Returns	Order-status composition, return/cancellation KPIs and trends
06 AI - Repurchase Prediction	Customer scores, opportunity segments, model metrics, map and prediction detail

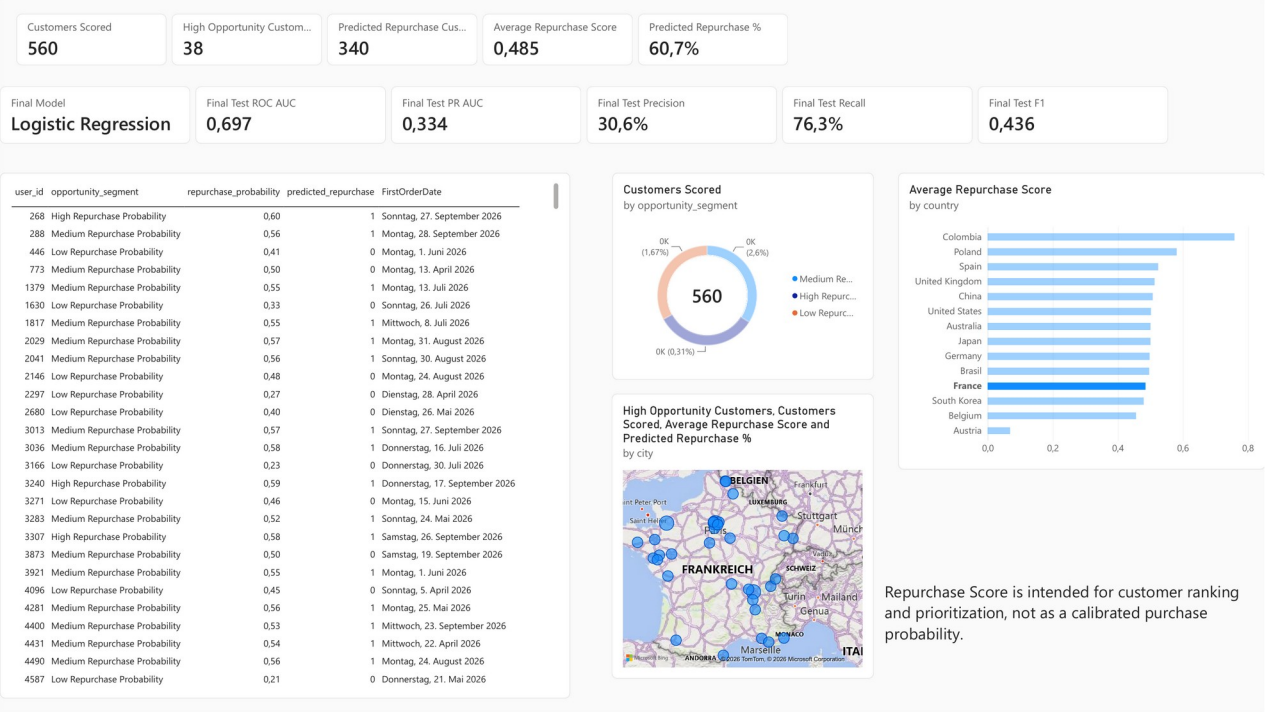


Figure 4. AI - Repurchase Prediction page. Power BI values are filter-context dependent; the Databricks notebook contains the validated full scoring population and model metrics.

8. Limitations and Interpretation

- TheLook is synthetic e-commerce data; business findings demonstrate methodology rather than a live commercial operation.
- The source data contains an artificial maximum number of orders per customer; the model therefore focuses only on first-to-second purchase.
- The positive target rate changes over calendar time, so production use would require drift monitoring, periodic retraining and recalibration.
- Weighted Logistic Regression scores overestimate absolute repurchase rates in some score ranges. The dashboard therefore labels the output as a Repurchase Score and uses it for ranking/prioritization.
- Power BI visuals depend on active filter context. Notebook row counts and model metrics are the reproducible reference for full-population validation.

9. Final Conclusion

The project demonstrates a complete BI lifecycle: public-source acquisition, scalable Databricks ingestion, data-quality controls, Silver cleansing, Gold dimensional modeling, financial KPI engineering, predictive modeling, model validation and Power BI delivery. The descriptive layer explains what happened across sales, customers, products and order outcomes; the predictive layer adds a defensible mechanism for prioritizing first-time customers for retention activity. The final solution is technically reproducible, analytically transparent and suitable for presentation as an end-to-end BI and ML project.

Evidence Used

This document was prepared from the submitted Databricks archive BI_Final_Project.dbc and the six-page Power BI export PROJECT.pdf. Numerical ML results are taken from the validated notebook run; visual BI results are taken from the exported Power BI report.